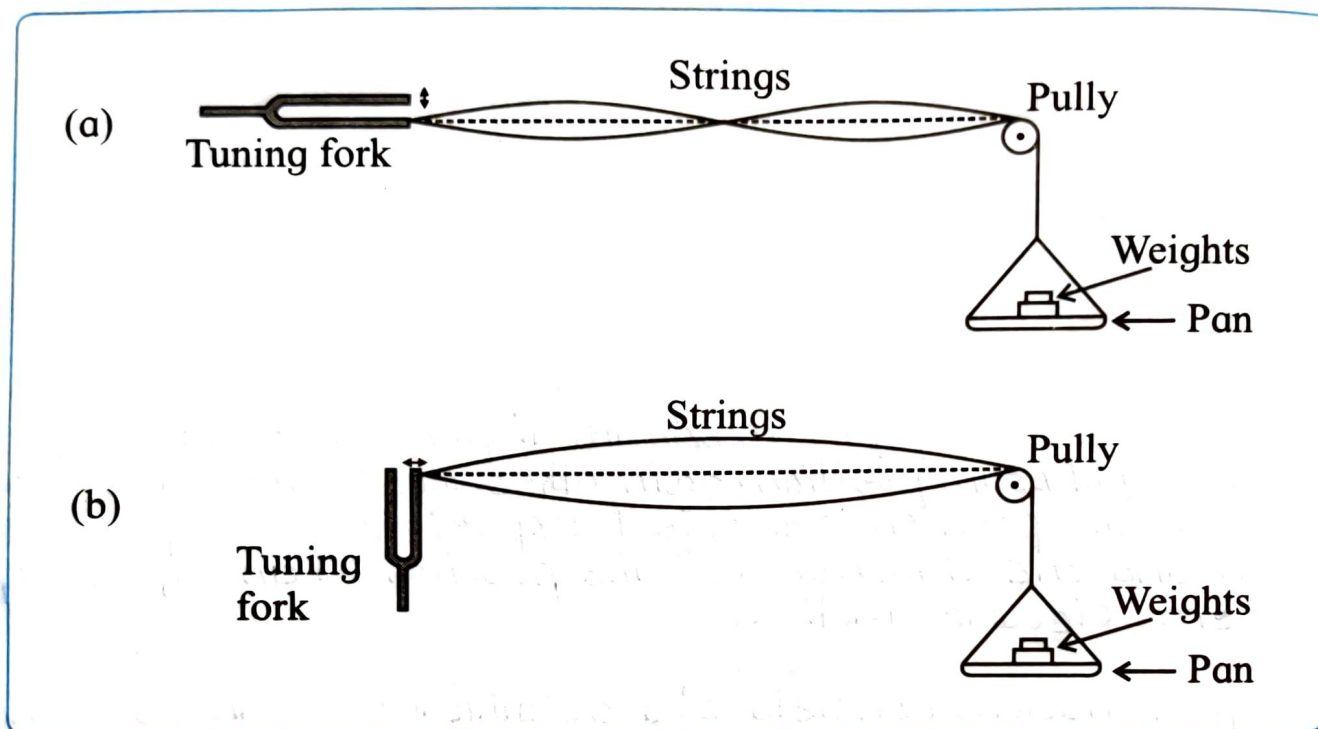


ACTIVITY NO. 3 MELDE'S EXPERIMENT

Aim: To demonstrate the formation of stationary waves and determine unknown frequency of a tuning fork by Melde's experiment.

Apparatus: A tuning fork, light flexible string, light pan, weight box, rubber, metre scale.

Diagram:



Procedure:

1. Tie one end of the string to one of the prongs of the tuning fork and the other end to a light pan.
2. Pass the string over the pulley. Adjust the pulley such that the string is horizontal.
3. Adjust the length of the string (between the tuning fork and pulley) to be one metre.
4. Adjust the tuning fork in the parallel/perpendicular position.
5. Keep the string under a suitable tension by adding weights to the pan.
6. Set the tuning fork vibrating by striking one of its prongs gently with a rubber hammer.
7. Adjust the tension (T) in the vibrating string such that a stable pattern of distinct loops are formed on the string. (Nodes should be well-defined.)
8. Count the number of loops (P) formed on the vibrating string.
9. Repeat the procedure to obtain different number of loops by changing the tension, keeping the length of the vibrating string constant throughout the experiment.
10. Calculate TP^2

Observations:

1. Mass of the pan = $M_0 = 5 \text{ g}$ (Weight = $M_0 = 4900 \text{ gwt}$)
2. Linear density (mass per unit length) of the string = $m = \dots = 0.0141 \text{ g/cm}$

Parallel position/Perpendicular position

Obs.no.	Weight added in the pan, M(gwt)	Tension T	Number of loops p	TP ²
1	10	9800	8	627200
2	15	14700	7	720300
3	20	19600	6	705600

$$\text{Mean TP}^2 = 684366.667$$

Calculations:

$$① p^2 = 8^2 = 64 \quad ② p^2 = 7^2 = 49 \quad ③ p^2 = 6^2 = 36$$

$$TP^2 = 9800 \times 64 = 627200$$

$$TP^2 = 14700 \times 49 = 720300$$

$$TP^2 = 19600 \times 36 = 705600$$

$$\frac{627200 + 720300 + 705600}{3} = 684366.667$$

Results:

$$TP^2 = 684366.667 \neq \text{constant.}$$

Precautions:

1. In determination of the linear density of the string, choose the string as long as possible so that its mass can be determined accurately.
2. Strike the tuning fork gently with a rubber hammer.
3. The pan should be light. It should not touch the table or stand.

Questions

1. What are the stationary waves?

Stationary (standing) waves are produced by the superposition of two waves of the same frequency and amplitude travelling in opposite directions.

2. Give some examples of stationary waves in real life.

Stationary waves are formed in stretched strings of musical instruments such as guitars, violins and pianos, and in air columns of instruments such as flutes, organ pipes and clarinets.

Remark and sign of teacher :